

GenMol UDLM engineering exploration

engineering-v12-ce-mask-prior | complete | 2026-09-07T12:56:25.414190+00:00

Engineering pilots only. Superiority has not been established.

Engineering pilots only: no superiority claim, final candidate lock, or final-seed evaluation.

Prospective training settings: initialization: Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.; optimizer updates per arm: 1000; global batch: 128; training seed: 1500; training GPUs: 2; requested example exposures per arm: 128000; common clean-target mask: All tokenizer special IDs immutable clean targets; full active corruption vocabulary.. The MDLM and paper baselines remain contextual comparisons.

Both prior arms use clean-token CE and the training settings recorded in this protocol. Changing the corruption prior changes reconstruction difficulty; equal updates and examples do not imply equal compute, and lower training loss does not establish better molecular generation.

Small samples and configuration selection create sampling uncertainty and selection bias; a higher pilot mean is not evidence of superiority.

Strict decoding uses fix=False. Released-compatible decoding uses SAFE repair and largest-component selection; report both because repairs can hide malformed outputs.

Validity and quality divide by requested samples. Uniqueness divides by valid samples. Quality counts unique molecules with QED $\geq$ 0.6 and SA $\leq$ 4. Diversity is the released PyTDC fingerprint metric over unique valid molecules.

Means and sample standard deviations average seeds equally within one configuration. Diversity is not pooled across seeds. Failed, pending, and invalid runs are excluded from means and remain disclosed.

Paper seeds are undisclosed and hardware differs. Local timings are descriptive; no speed claim relative to the paper is made.

Predeclared contrasts subtract empirical CE from MASK-rich CE within the same seed and decoding branch. Every declared contrast is retained, including negative outcomes. Paired mean and sample SD are withheld unless all declared seed pairs define that metric; unavailable pairs remain visible. Sample SD describes seed spread, not a confidence interval. Pairing seed labels does not imply coupled molecular trajectories.

Prospective sampling design

```
{"corrector_steps_per_molecule": 0, "inference_eps": 1e-05, "inference_weights": "ema", "min_add_len": 40, "parameterization_controls": "Both CE arms convert clean-denoiser logits to LOO at current state/time before temperature and top-p; no extra model call.", "predictor_transitions_per_molecule": 128, "prior_comparison": {"primary": {"control_config": "e_ce_t100", "temperature": 1.0, "treatment_config": "mask_ce_t100"}, "secondary": {"control_config": "e_ce_t050", "temperature": 0.5, "treatment_config": "mask_ce_t050"}}, "randomness": 0.0, "raw_loo_top_p": 1.0}
```

V9/V10 and the frozen MDLM transfer diagnostic informed this exploratory prior comparison; no superiority or independent confirmation claim.

Both arms use clean CE; changed corruption and reverse laws change reconstruction difficulty. Equal updates/exposures do not imply equal runtime.

Original V11 failed before training and remains failed. V11b is a separate successful fresh attempt; original V8 CT remains failed with a separate checkpoint audit and is context only.

Two seeds of 100 requests are small. Pair by seed, not molecular trajectory. Report every configuration and failure, both decoding branches, all four metrics and signed MASK-minus-empirical contrasts.

Quality counts unique valid QED $\geq$ 0.6 and SA $\leq$ 4 molecules divided by all requests. Report recovery and largest-component selection frequencies with denominators.

Disclose final editable MASK counts and their editable-position denominator from summary sampled_token_control_audit; decoded strings skip specials and cannot supply this count.

128000 configured training exposures are not necessarily distinct molecules. Finite final checkpoints do not establish finiteness of every update.

Local MDLM .858 and paper GenMol V1 .846 use different training/sample budgets and are context only; no automatic promotion or extra training. Final seeds0/1/2 remain reserved.

Prior comparison

Both arms predict clean-denoiser probabilities using CE training, with empirical versus MASK-rich empirical stationary priors. Both convert to LOO logits at the current noisy state and time before temperature or top-p, with no extra backbone evaluation. Changing the prior changes the training corruption and reverse law. Checkpoint evidence below describes completed runs only; the prospective design also covers pending runs.

Training provenance: {"accumulate_grad_batches": 4, "arms": {"E_CE": {"checkpoint_sha256": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "checkpoint_size_bytes": 1636492999, "controller_status": "completed", "launch_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/launch_manifest.json", "sha256": "903819961be96f71da79774d723626d8c4eb1aa50bf482d2a644c7a0dda22315", "size_bytes": 17020}, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "protocol": {"relative_path": "experiments/udlm/protocols/engineering_v8b_ce_followup.json", "sha256": "acb286ac4b938aa049327740affd2dea5beb0ded745cd51dca627f5e8fd9de8b", "size_bytes": 3161}, "request_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/request_manifest.json", "sha256": "c05fafdd1472e894b553afe580b051b1b2eec4c5dd47f7323dfb939e51f958d", "size_bytes": 8621}, "resolved_config_sha256": "80bd9866ef157690e0ef9ff553b0346b3b91f989c0193d0484c4997ac089b2bb", "source_revision": "b84f21ba1f30ceb62d2626daf2caad8adb6534cf", "terminal_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/terminal_manifest.json", "sha256": "2dd0423257e8908548b69a28b8aa24b3cc956507944c343e24ef615253af2205", "size_bytes": 11719}, "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}, "udlm_prior_metadata": {"active_token_ids_sha256": "2fd62050919fdf6ee578df881733e47834f662406c219a1f247116e04396a5ae", "active_vocab_size": 1880, "antithetic_sampling": true, "comparison_role": "empirical_prior_treatment", "excluded_token_ids": [], "frequency_active_token_count": 517090, "frequency_artifact_path": "experiments/udlm/token_frequency/train_first_10000.json", "frequency_artifact_schema_version": 1, "frequency_artifact_sha256": "088c78e75611f3cc42c4011e1da6f65a377e673b9cba07a28b126b0fc62f06ed", "frequency_content_token_count": 517090, "frequency_dataset_repo_id": "datamol-io/safe-gpt", "frequency_dataset_revision": "b83175cd7394e7a4027478a35b2f9d1dda3ac62f", "frequency_dataset_selection": "first 10000 streaming rows", "frequency_dataset_split": "train", "frequency_example_count": 10000, "frequency_implementation_git_sha": "56a96b2cd02f9be648a641c51c3d2d8b1ff3033b", "frequency_ordered_text_sha256": "53aee8e5592fc96159788e86519abbbcc9f1ab7c6348a1cb59a939bd57051d8f", "full_vocab_size": 1880, "noise_eps": 0.001, "objective_scope": "model_dependent_ct_integrand_without_parameter_independent_endpoint_kl", "prior_source": "pinned_frequency_artifact_uniform_mixture", "process_family": "rank_one_continuous_categorical", "sampling_eps": 0.001, "schedule_variant": "schedule_consistent_residual_forward_and_loss", "schema_version": 1, "stationary_probs_sha256": "3b6ce79449e0abc083bc860b3543be554667435ce9f7f1b92c0f960ff27400c3", "tokenizer_json_sha256": "0db5f4dbdc7e8ff759e98483759611a426e187ee7f3f0a91edc8800abe7bf140", "tokenizer_repo_id": "datamol-io/safe-gpt", "tokenizer_revision": "3d5fa0988383e898d5ac5db7cd52bf715bc37061", "uniform_mixture_weight": 0.0002, "variant": "empirical_frequency"}}, {"MASK_CE": {"checkpoint_sha256": "62299de99d8c003e3776215efce9643cca196091a6284f351de2b404a22c2e8d", "checkpoint_size_bytes": 1636493702, "controller_status": "completed", "launch_receipt": {"relative_path": "output/udlm/engineering_v11b/mask_ce_1000_b128_w2/launch_manifest.json", "sha256": "457912bcf1975f0adb72cb2b5f02ec07c32915dc2119f184c19d6003edb60f85", "size_bytes": 24826}, "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "protocol": {"relative_path": "experiments/udlm/protocols/engineering_v11b_mask_prior.json", "sha256": "5ecf638fa8fc7a3707a0497c8a358697610f52c8af7f2d6468458890e26368eb", "size_bytes": 7982}, "request_receipt": {"relative_path": "output/udlm/engineering_v11b/mask_ce_1000_b128_w2/request_manifest.json", "sha256": "a039045f0e3589b3c15ac56fa215cb27bf0bde36b87902c0e1028c456b45f956", "size_bytes": 14003}, "resolved_config_sha256":

"5d73af507e81b643cf8f35225ab5d796357024d1665d039fdf40b5f37df15698", "source_revision": "be18a3244a717978e027d6e43ba0ac20b8137e4a", "terminal_receipt": {"relative_path": "output/udlm/engineering_v11b/mask_ce_1000_b128_w2/terminal_manifest.json", "sha256": "891f5d6d609146a02ebb18caa7176bf555e3f67fe2493795c8a3e1fccb264779", "size_bytes": 17667}, "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}, "udlm_prior_metadata": {"active_token_ids_sha256": "2fd62050919fdf6ee578df881733e47834f662406c219a1f247116e04396a5ae", "active_vocab_size": 1880, "antithetic_sampling": true, "base_stationary_probs_sha256": "3b6ce79449e0abc083bc860b3543be554667435ce9f7f1b92c0f960ff27400c3", "comparison_role": "mask_rich_empirical_prior_treatment", "excluded_token_ids": [], "frequency_active_token_count": 517090, "frequency_artifact_path": "experiments/udlm/token_frequency/train_first_10000.json", "frequency_artifact_schema_version": 1, "frequency_artifact_sha256": "088c78e75611f3cc42c4011e1da6f65a377e673b9cba07a28b126b0fc62f06ed", "frequency_content_token_count": 517090, "frequency_dataset_repo_id": "datamol-io/safe-gpt", "frequency_dataset_revision": "b83175cd7394e7a4027478a35b2f9d1dda3ac62f", "frequency_dataset_selection": "first 10000 streaming rows", "frequency_dataset_split": "train", "frequency_example_count": 10000, "frequency_implementation_git_sha": "56a96b2cd02f9be648a641c51c3d2d8b1ff3033b", "frequency_ordered_text_sha256": "53aee8e5592fc96159788e86519abbbcc9f1ab7c6348a1cb59a939bd57051d8f", "full_vocab_size": 1880, "mask_mixture_weight": 0.9, "mask_token_id": 4, "noise_eps": 0.001, "objective_scope": "model_dependent_ct_integrand_without_parameter_independent_endpoint_kl", "prior_source": "mask_point_mass_mixture_with_pinned_smoothed_frequency", "process_family": "rank_one_continuous_categorical", "sampling_eps": 0.001, "schedule_variant": "schedule_consistent_residual_forward_and_loss", "schema_version": 1, "stationary_probs_sha256": "07d011c07bdac8fdb096f514ded6e2d1b6850268c48bec8dd0e3f3853d013eb", "tokenizer_json_sha256": "0db5f4dbdc7e8ff759e98483759611a426e187ee7f3f0a91edc8800abe7bf140", "tokenizer_repo_id": "datamol-io/safe-gpt", "tokenizer_revision": "3d5fa0988383e898d5ac5db7cd52bf715bc37061", "uniform_mixture_weight": 0.0002, "variant": "mask_rich_empirical"}}, "common_mask_policy": "All tokenizer special IDs immutable clean targets; full active corruption vocabulary.", "conditioning_variant": "film_adaln", "example_exposures_per_arm": 128000, "global_batch_size": 128, "gpu_count": 2, "initialization": "Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.", "matched_common_config_definition": "Empirical-arm configuration after removing parameterization and normalizing callback.dirpath. The MASK arm matches after restoring empirical_frequency and removing mask_mixture_weight.", "matched_common_config_sha256": "f373121441fca1aaa2f2657a76d7f7ce66be199cd35637597d0df240e257e89a", "micro_batch_size": 16, "optimizer_updates": 1000, "original_failed_campaign": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/campaign_terminal.json", "sha256": "98075a11c7340dafcc855887348201477981bd2a1a8f5d39053da24866cc663b", "size_bytes": 846}, "original_v11_failure": {"relative_path": "output/udlm/engineering_v11/mask_ce_1000_b128_w2/terminal_manifest.json", "sha256": "6a4aa47e1f7b76ad5efc3ce03cd3e4a55c0db4d95778b0cf60cc55cab7207408", "size_bytes": 14510}, "original_v8_ct_context": {"checkpoint_sha256": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "checkpoint_status": "separately_validated_after_controller_failure", "controller_status": "failed", "post_exit_audit": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/post_exit_audit/post_exit_audit.json", "sha256": "959151c37072431be23b1f5696d7215da5291e0bace40d6f307864a7262f93c2", "size_bytes": 9166}, "resolved_config_sha256": "adb518da62ea14d0e0b2e11f1ea91e058b25d0d35a59c3e8edfef060b5f0c46", "source_revision": "c8434dda105c5fb2a15bb784d24e0387062c733e", "terminal_receipt": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/ct/terminal_manifest.json", "sha256": "1021ba513564a25ca26ea3c2d6c148d271e1cd98d27078056285e848c9c69b5e", "size_bytes": 9822}}, "original_v8_protocol": {"relative_path": "experiments/udlm/protocols/engineering_v8_objectives.json", "sha256": "a27875752d25a4a7928401434f9bd90ee6e9d01539586bab70ce3ba75499026e"}, "peak_learning_rate": 0.0003, "scheduler": {"decay_floor_lr": 3e-06, "horizon_updates": 1000, "name": "half_cosine_with_linear_warmup_and_floor", "warmup_updates": 50}, "training_seed": 1500, "uniform_mixture": 0.0002}

Reference means (context only)

Reference	validity	uniqueness	quality	diversity
local_mdml_50000	1.000000	0.998667	0.858000	0.823021
paper_genmol_v1_table1	1.000000	0.997000	0.846000	0.818000

Sampling evaluation budget

Counts are per molecule. Each predictor transition and each fresh Gibbs correction consumes one backbone evaluation. Predictor-only counts equal recorded NFE; missing runs have no observed budget.

Config	Seed	Total NFE	Predictors	Correctors
e_ce_t100	2000	128	128	0
e_ce_t100	2001	128	128	0
mask_ce_t100	2000	128	128	0
mask_ce_t100	2001	128	128	0
e_ce_t050	2000	128	128	0
e_ce_t050	2001	128	128	0
mask_ce_t050	2000	128	128	0
mask_ce_t050	2001	128	128	0

Checkpoint logit interpretation

Config	Seed	Objective	Parameterization	Weights
e_ce_t100	2000	clean CE	x0_denoiser	ema
e_ce_t100	2001	clean CE	x0_denoiser	ema
mask_ce_t100	2000	clean CE	x0_denoiser	ema
mask_ce_t100	2001	clean CE	x0_denoiser	ema
e_ce_t050	2000	clean CE	x0_denoiser	ema
e_ce_t050	2001	clean CE	x0_denoiser	ema
mask_ce_t050	2000	clean CE	x0_denoiser	ema
mask_ce_t050	2001	clean CE	x0_denoiser	ema

Per-configuration mean, equal seed weights

released_comparable

Config	Seeds complete	validity	uniqueness	quality	diversity
e_ce_t100	2/2	0.9900 +/- 0.0141	0.9950 +/- 0.0071	0.4250 +/- 0.0636	0.9008 +/- 0.0034
mask_ce_t100	2/2	0.9950 +/- 0.0071	0.9950 +/- 0.0071	0.5000 +/- 0.0566	0.9046 +/- 0.0004
e_ce_t050	2/2	0.9800 +/- 0.0141	1.0000 +/- 0.0000	0.4850 +/- 0.0212	0.8945 +/- 0.0074
mask_ce_t050	2/2	0.9950 +/- 0.0071	0.9949 +/- 0.0071	0.5400 +/- 0.0283	0.8863 +/- 0.0042

strict

Config	Seeds complete	validity	uniqueness	quality	diversity
e_ce_t100	2/2	0.5850 +/- 0.0636	1.0000 +/- 0.0000	0.2750 +/- 0.0212	0.8854 +/- 0.0061
mask_ce_t100	2/2	0.5850 +/- 0.0354	1.0000 +/- 0.0000	0.3400 +/- 0.0566	0.8823 +/- 0.0007
e_ce_t050	2/2	0.7150 +/- 0.0071	1.0000 +/- 0.0000	0.3750 +/- 0.0354	0.8830 +/- 0.0079
mask_ce_t050	2/2	0.7650 +/- 0.0495	1.0000 +/- 0.0000	0.4200 +/- 0.0424	0.8765 +/- 0.0003

Predeclared paired MASK minus empirical contrasts

Differences use the same seed and decoding branch, with equal seed weights. Values are on the metric's 0-to-1 scale: 0.01 is one percentage point for quality. All declared pairs must define a metric before its mean and sample SD are shown. SD is seed spread, not a confidence interval. Seed labels do not imply paired molecules.

primary | temperature 1.0: mask_ce_t100 minus e_ce_t100

released_comparable

Metric	Seed	Empirical CE	MASK CE	MASK minus empirical	Pair status
validity	2000	+1.000000	+1.000000	+0.000000	completed
validity	2001	+0.980000	+0.990000	+0.010000	completed
uniqueness	2000	+0.990000	+0.990000	+0.000000	completed
uniqueness	2001	+1.000000	+1.000000	+0.000000	completed
quality	2000	+0.380000	+0.540000	+0.160000	completed

Metric	Seed	Empirical CE	MASK CE	MASK minus empirical	Pair status
quality	2001	+0.470000	+0.460000	-0.010000	completed
diversity	2000	+0.898434	+0.904817	+0.006383	completed
diversity	2001	+0.903207	+0.904317	+0.001110	completed

Metric	Pairs defined	Mean MASK minus empirical	Sample SD
validity	2/2	+0.005000	0.007071
uniqueness	2/2	+0.000000	0.000000
quality	2/2	+0.075000	0.120208
diversity	2/2	+0.003746	0.003729

strict

Metric	Seed	Empirical CE	MASK CE	MASK minus empirical	Pair status
validity	2000	+0.630000	+0.610000	-0.020000	completed
validity	2001	+0.540000	+0.560000	+0.020000	completed
uniqueness	2000	+1.000000	+1.000000	+0.000000	completed
uniqueness	2001	+1.000000	+1.000000	+0.000000	completed
quality	2000	+0.260000	+0.380000	+0.120000	completed
quality	2001	+0.290000	+0.300000	+0.010000	completed
diversity	2000	+0.881115	+0.881820	+0.000705	completed
diversity	2001	+0.889762	+0.882860	-0.006902	completed

Metric	Pairs defined	Mean MASK minus empirical	Sample SD
validity	2/2	+0.000000	0.028284
uniqueness	2/2	+0.000000	0.000000
quality	2/2	+0.065000	0.077782
diversity	2/2	-0.003099	0.005379

secondary | temperature 0.5: mask_ce_t050 minus e_ce_t050

released_comparable

Metric	Seed	Empirical CE	MASK CE	MASK minus empirical	Pair status
validity	2000	+0.990000	+1.000000	+0.010000	completed
validity	2001	+0.970000	+0.990000	+0.020000	completed
uniqueness	2000	+1.000000	+1.000000	+0.000000	completed
uniqueness	2001	+1.000000	+0.989899	-0.010101	completed
quality	2000	+0.500000	+0.560000	+0.060000	completed
quality	2001	+0.470000	+0.520000	+0.050000	completed
diversity	2000	+0.889283	+0.883360	-0.005924	completed
diversity	2001	+0.899687	+0.889340	-0.010346	completed

Metric	Pairs defined	Mean MASK minus empirical	Sample SD
validity	2/2	+0.015000	0.007071
uniqueness	2/2	-0.005051	0.007142
quality	2/2	+0.055000	0.007071
diversity	2/2	-0.008135	0.003127

strict

Metric	Seed	Empirical CE	MASK CE	MASK minus empirical	Pair status
validity	2000	+0.720000	+0.800000	+0.080000	completed
validity	2001	+0.710000	+0.730000	+0.020000	completed
uniqueness	2000	+1.000000	+1.000000	+0.000000	completed
uniqueness	2001	+1.000000	+1.000000	+0.000000	completed
quality	2000	+0.400000	+0.450000	+0.050000	completed

Metric	Seed	Empirical CE	MASK CE	MASK minus empirical	Pair status
quality	2001	+0.350000	+0.390000	+0.040000	completed
diversity	2000	+0.877424	+0.876253	-0.001171	completed
diversity	2001	+0.888618	+0.876684	-0.011935	completed

Metric	Pairs defined	Mean MASK minus empirical	Sample SD
validity	2/2	+0.050000	0.042426
uniqueness	2/2	+0.000000	0.000000
quality	2/2	+0.045000	0.007071
diversity	2/2	-0.006553	0.007611

Every scheduled seed

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
e_ce_t100	2000	completed	100	100	0.3800	0.2600	15.24
e_ce_t100	2001	completed	100	100	0.4700	0.2900	15.45
mask_ce_t100	2000	completed	100	100	0.5400	0.3800	14.85
mask_ce_t100	2001	completed	100	100	0.4600	0.3000	15.63
e_ce_t050	2000	completed	100	100	0.5000	0.4000	14.87
e_ce_t050	2001	completed	100	100	0.4700	0.3500	15.11
mask_ce_t050	2000	completed	100	100	0.5600	0.4500	15.24
mask_ce_t050	2001	completed	100	100	0.5200	0.3900	15.38

Preserved v4 diagnostic failure

V4 remains campaign_incomplete; its failed D attempt is never reclassified or reused for ranking.

completion validation failed: Seed 1100 has benchmark artifacts that do not match the requested run or fail integrity checks: - config.effective={'device': 'cuda:0', 'diffusion_type': 'udlm', 'exclude_special_tokens': False, 'inference_eps': 1e-05, 'min_add_len': 40, 'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32,

'num_steps': 128, 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'prior_variant': 'empirical_frequency', 'randomness': 0.0, 'raw_loo_top_p': 1.0, 'softmax_temp': 1.0}; expected {'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'diffusion_type': 'udlm', 'softmax_temp': 1.0, 'randomness': 0.0, 'min_add_len': 40, 'num_steps': 128, 'inference_eps': 1e-05, 'exclude_special_tokens': False, 'prior_variant': 'empirical_frequency', 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'device': 'cuda:0'} - config.effective_sha256='278d4476e98bc0b5905e16c2158dd75a3a5a85250e4a48726640afc0e5529608'; expected '3ee30748fa7d7fcc675fde0fe1e8ff15f23fdc3374ba76eed03466ba306ec02f' Refusing to skip or relaunch into the existing directory. Inspect the artifacts or choose a fresh output root.

Configuration and artifact provenance

Protocol SHA-256: f4ea756fabb9e5133c72f302e8398bcbb8d0eabaa81f183b017af8c19b02cadd

v12-e-ce-t100 / seed 2000: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1842}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1865, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": 3}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v12/v12-e-ce-t100/seed_2000/raw_samples.csv", "sha256": "2f7aeed34d62a4ce8285287468400755d6300e73a6262c00e4e117a72e5701e9", "size_bytes": 32935}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c61a1ef4b42fd2f8aba817e5e3e3554afc64eaa1", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v12-e-ce-t100 / seed 2001: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true,

"single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1842}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1865, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": 3}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v12/v12-e-ce-t100/seed_2001/raw_samples.csv", "sha256": "473c9897f2f72f9b7c93d2d5225c032e82c6c5e9855851fca6654b1fbc8c1b18", "size_bytes": 31685}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c61a1ef4b42fd2f8aba817e5e3e3554afc64eaa1", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v12-mask-ce-t100 / seed 2000: {"checkpoint": "62299de99d8c003e3776215efce9643cca196091a6284f351de2b404a22c2e8d", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1842}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1865, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": 3}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v12/v12-mask-ce-t100/seed_2000/raw_samples.csv", "sha256": "7f618bc2e4f2424bc675a5dfa331a83b934935310e88ba7771899e16e3fe60ce", "size_bytes": 31923}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c61a1ef4b42fd2f8aba817e5e3e3554afc64eaa1", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v12-mask-ce-t100 / seed 2001: {"checkpoint": "62299de99d8c003e3776215efce9643cca196091a6284f351de2b404a22c2e8d", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true,

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v12-e-ce-t050 / seed 2000: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema", "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1842}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1865, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": 3}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema", "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v12/v12-e-ce-t050/seed_2000/raw_samples.csv", "sha256": "252d36cfd849291b531924732e2a48f7e4bf2e375229797b76ff3a78b15d1095", "size_bytes": 32840}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c61a1ef4b42fd2f8aba817e5e3e3554afc64eaa1", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v12-e-ce-t050 / seed 2001: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema", "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true,

"single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1564}, {"pid": 623773, "process_name": "/home/aidar.alimbayev/Documents/genmolv2/run_sources/running_mean_repair/.venv/bin/python", "used_memory_mib": 2154}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 3747, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": 5}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v12/v12-e-ce-t050/seed_2001/raw_samples.csv", "sha256": "423c7f380826ad0a36110e7f34cebc793981ff57a762f82c8c7f9d8db229589e", "size_bytes": 32663}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c61a1ef4b42fd2f8aba817e5e3e3554afc64eaa1", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v12-mask-ce-t050 / seed 2000: {"checkpoint": "62299de99d8c003e3776215efce9643cca196091a6284f351de2b404a22c2e8d", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1842}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1865, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": 3}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v12/v12-mask-ce-t050/seed_2000/raw_samples.csv", "sha256": "2f7d7fc4d58888233de754024bc7d45cecb7ffa48d07747ac09c787ebcc884ce", "size_bytes": 33600}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c61a1ef4b42fd2f8aba817e5e3e3554afc64eaa1", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

v12-mask-ce-t050 / seed 2001: {"checkpoint": "62299de99d8c003e3776215efce9643cca196091a6284f351de2b404a22c2e8d", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "4e1febe2684beebdbf3d4c86aa01bc24ed1d3941af67b63473979e2ae810ee45", "prior_variant": "mask_rich_empirical", "randomness": 0.0,

"randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1564}, {"pid": 623773, "process_name": "/home/aidar.alimbayev/Documents/genmolv2/run_sources/running_mean_repair/.venv/bin/python", "used_memory_mib": 2154}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 3747, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": 5}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v12/v12-mask-ce-t050/seed_2001/raw_samples.csv", "sha256": "c3790a6acb057432763d7b2d9ea1a09c2634f31a3338462a145399474c38f7a6", "size_bytes": 33084}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "c61a1ef4b42fd2f8aba817e5e3e3554afc64eaa1", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}